Satyabrata Das

Software Engineer | Backend, Full-Stack & Machine Learning Systems
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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building production backend services, real-time systems, full-stack applications, and machine learning pipelines. Proficient in Python, C++, TypeScript, SQL, RESTful APIs, AWS, Docker, and CI/CD, with strong foundations in data structures, algorithms, object-oriented design, and concurrency. Track record of owning features from design to deployment and delivering measurable gains in latency, reliability, and load time for products serving 20K+ users. Pursuing an M.S. in Artificial Intelligence Systems at the University of Florida.

KEY SKILLS

- **Languages:** Python, C++, TypeScript, JavaScript, SQL, Swift
- **Backend & APIs:** RESTful APIs, FastAPI, Node.js, Express.js, WebSockets, Real-time Systems, ETL Pipelines
- **Cloud & Databases:** AWS, GCP, PostgreSQL, MySQL, MongoDB, SQL & NoSQL Databases
- **Frontend, Mobile & DevOps:** React, React Native, iOS (Core Data, GCD), Docker, CI/CD, Git, Linux, Grafana
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, LLM APIs, RAG, Pandas

PROFESSIONAL EXPERIENCE

Evy | *Software Engineer, Machine Learning* **May 2026 – Present**
- Owned the first production release of a real-time AI interview monitoring platform end to end, from design to deployment.
- Optimized the in-browser inference and deployment pipeline to sustain 40 FPS on consumer webcams and low-end hardware.
- Strengthened reliability under glare, head pose, and low light via adaptive thresholding across 50 test configurations.
- Developed PyTorch/TensorFlow anomaly detection models over 3 real-time signal streams (gaze, iris, blink).

University of Florida | *Machine Learning Research Assistant* | Gainesville, FL **Mar 2026 – May 2026**
- Created an end-to-end Python data pipeline running MediaPipe 3D pose estimation across 1,000+ clinical patient videos.
- Extracted 3 biomechanical feature sets (joint angles, angular velocity, symmetry) from landmark time-series data.
- Trained regression models on 1,000+ recordings to predict clinician-rated severity scores, automating manual review.

SouthEnd Psychiatry | *IT & Operations Contractor* **Sep 2024 – Aug 2025**
- Automated intake reconciliation with Python scripts, eliminating 30 hours of manual data entry per week.
- Reduced scheduling conflicts by 25% across 5,000+ monthly appointments through automated data validation.
- Maintained EHR systems for 20 clinicians and 10,000+ patient records under strict HIPAA compliance.

ARC Document Solutions | *Software Engineer* **Jul 2022 – Sep 2024**
- Engineered a real-time WebSocket sync layer for 20K+ MAU, reducing latency 25% and support tickets ~30% QoQ.
- Integrated 5 payment gateway APIs (Razorpay, Paytm, Cashfree, Braintree, Elavon), cutting failed transactions 15%.
- Redesigned the data and concurrency layer with GCD multithreading, cutting load time 30% and eliminating UI freezes.

SCI-BI Software Pvt Ltd | *Software Engineer Trainee* **Jul 2021 – Dec 2021**
- Delivered Python and SSIS ETL data pipelines processing 10M+ records daily with 99%+ data accuracy.
- Streamlined recurring data workflows, saving 20 hours of manual work per week.
- Tuned SQL queries with Prometheus/Grafana monitoring, cutting report time 40% and pipeline failures 25%.

PROJECTS

bendwith.us: AI Physical-Therapy Coach Live
- Built an on-device pose-tracking rehab coach (MediaPipe, React, Capacitor) for 5 joints across 3 platforms, with 130+ bilingual ElevenLabs voice cues and 10 Hz session replay for therapists.
- Engineered a FastAPI backend with WebSocket/SSE streaming, storing 100 ms angle data in TimescaleDB (∼11:1 compression) and using Gemini to draft clinician-approved plan changes.

Intelligent Bistro — *React Native, TypeScript, Node.js, Express.js, Gemini API* GitHub
- Architected a full-stack app where an LLM maps orders into JSON cart actions, with rule-based fallback for fault tolerance.

JuggleIQ — *Python, FastAPI, React, YOLO, MediaPipe* GitHub
- Built a GPU-backed FastAPI service for video detection and tracking, with a React dashboard and annotated replay.

SummarIQ — *Python, FastAPI, FAISS, HuggingFace Transformers, LangChain* GitHub
- Deployed a FastAPI semantic search API over 200+ documents at sub-1.4s latency with FAISS, improving accuracy ~18%.

EDUCATION

University of Florida Gainesville, FL
Master of Science, Artificial Intelligence Systems Aug 2025 – May 2027 (Expected)
- Coursework: AI Systems, Applied Deep Learning, Machine Learning for AI, Computer Vision & Image Processing

Guru Nanak Institute of Technology Kolkata, India
Bachelor of Technology, Computer Science; GPA: 3.80/4.0 Jul 2018 – Jun 2022
- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Cyber Security